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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (Civil/Mechanical) (Credit System) Degree Examinations **Supplementary Examinations - December 2025**

21CV301 / 20CV301 / 21ME301 / 20ME301 - ENGINEERING MATHEMATICS - III

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I

	Marks	BT*	CO*	PO*
1. a) If $r = \sqrt{x^2 + y^2 + z^2}$, find ∇r^n and hence show that $\nabla^2 r^n = n(n+1)r^{n-2}$	7	L*1	1	1
b) Find the work done in moving a particle in the force field $\vec{F} = 3x^2 \hat{i} + (2xz - y) \hat{j} + z \hat{k}$ along the straight line from (0,0,0) to (2,1,3).	7	L3	2	2
c) Show that the vector field $\vec{F} = (y^2 \cos x + z^3) \hat{i} + (2y \sin x - 4) \hat{j} + (3xz^2 + 2) \hat{k}$ is conservative. Find its scalar potential ϕ .	6	L3	1	2
2. a) State and prove Green's theorem in plane.	7	L1	2	1
b) A particle moves along the curve $x = 1 - t^3, y = 1 + t^2, z = 2t - 5$ where t is the time variable. Find the components of velocity and acceleration at $t=2$ in the direction given by $2\hat{i} + \hat{j} + 2\hat{k}$.	7	L3	1	2
c) In what direction from (3, -1, 2) is the directional derivative of $\phi = \log(x^2 + y^2 + z^2)$ maximum? Find also the magnitude of this maximum.	6	L3	1	2
3. a) i) Define solenoidal and irrotational vector functions. (ii) Show that $\vec{A} = (y + z)\hat{i} + (z + x)\hat{j} + (x + y)\hat{k}$ is irrotational.	7	L3	1	2
b) Prove that the spherical co-ordinate system is an orthogonal curvilinear coordinate system.	7	L1	1	1
c) State divergence theorem of Gauss and use it to evaluate $\int_S \vec{F} \cdot \hat{n} \, ds$ if $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is the surface of the cube bounded $x=0, x=1, y=0, y=1, z=0$ and $z=1$.	6	L3	2	2

Unit – II

4. a) If $f(z) = u(x, y) + i v(x, y)$ is analytic in a domain D , then prove that at any point of D , $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.	7	L1	3	1
b) Find the bilinear transformation which maps $z_1 = 0, z_2 = 1, z_3 = \infty$ onto $w_1 = -1, w_2 = -i, w_3 = 1$.	7	L3	4	2
c) Write the Laurent's expansion of $\frac{1}{z^2 - 3z + 2}$ in the region (i) $ z < 1$ (ii) $ z > 2$.	6	L3	4	2

P.T.O.

5. a) State and prove Cauchy's Integral theorem. 7 L1 4 1
 b) State the residue theorem and use it to evaluate $\int_C \frac{(3z^3+2) dz}{(z-1)(z+3i)(z-3i)}$ if C is the circle $|z-1| = 2$ 7 L3 4 2
 c) Show that the function $f(z) = \sin z$ is analytic and hence obtain derivative of this function. 6 L3 3 2
6. a) Using Cauchy's Integral formula, evaluate $\int_C \frac{e^{2z}}{(z+1)^4} dz$ if C is the circle $|z| = 2$ 7 L3 4 2
 b) Determine the analytic function $f(z) = u + i v$. Given that $v = \log(x^2 + y^2) + x - 2y$ 7 L3 3 2
 c) Discuss the transformation of $w = \sin z$. 6 L1 3 1

Unit – III

7. a) Find the Fourier series of $f(x) = x^2$, $-\pi < x < \pi$, $f(x+2\pi) = f(x)$. 7 L3 5 2
 b) Find a_0, a_1, b_1 of the Fourier series of the following data.
- | | | | | | | | |
|---|-----|-----------------|------------------|-------|------------------|------------------|--------|
| x | 0 | $\frac{\pi}{3}$ | $\frac{2\pi}{3}$ | π | $\frac{4\pi}{3}$ | $\frac{5\pi}{3}$ | 2π |
| y | 7.9 | 7.2 | 3.6 | 0.5 | 0.9 | 6.8 | 7.9 |
- c) Find the half range Fourier sine series of $f(x) = x$, $0 < x < 2$. 7 L3 5 2
 6 L3 5 2
8. a) Find the Fourier sine transform of $\frac{e^{-ax}}{x}$. 7 L3 5 2
 b) State and prove convolution theorem for Fourier transforms. 7 L1 5 1
 c) Find the inverse Fourier transform of $F(s) = e^{-a|s|}$ 6 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
